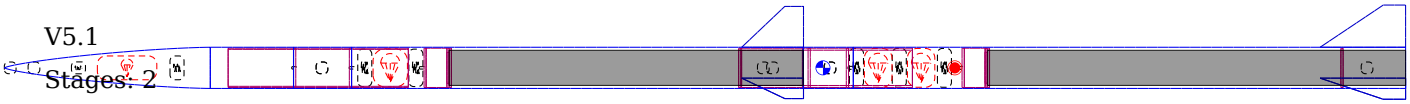
Rocket Design



V5.1

Stages: 2

Mass (with motors): 19245 g

Stability: 3.2 cal / 9.41 %

CG: 175 cm

CP: 203 cm

Nominal

Altitude	14159 m
Flight Time	812 s
Time to Apogee	53.4 s
Optimum Delay	35.8 s
Velocity off Pad	39 m/s
Max Velocity	754 m/s
Velocity at Deployment	19.7 m/s
Landing Velocity	12.6 m/s

Motor	Avg Thrust	Burn Time	Max Thrust	Total Impulse	Thrust to Wt	Motor Wt	Size
5880M 1160-P	1154 N	5.1 s	1361 N	5886 Ns	12.52:1	3454 g	75/757 mm
8429M 2020-P	2021 N	4.17 s	2649 N	8429 Ns	10.71:1	4349 g	75/893 mm
Total:				14314 Ns	10.71:1	7803 g	

Main

Altitude	14468 m
Flight Time	1670 s
Time to Apogee	54.1 s
Optimum Delay	36.5 s
Velocity off Pad	38.9 m/s
Max Velocity	754 m/s
Velocity at Deployment	78.2 m/s
Landing Velocity	12.8 m/s

Motor	Avg Thrust	Burn Time	Max Thrust	Total Impulse	Thrust to Wt	Motor Wt	Size
5880M 1160-P	1154 N	5.1 s	1361 N	5886 Ns	12.52:1	3454 g	75/757 mm
8429M 2020-P	2021 N	4.17 s	2649 N	8429 Ns	10.71:1	4349 g	75/893 mm
Total:				14314 Ns	10.71:1	7803 g	

Ballistic

Altitude	14469 m
Flight Time	127 s
Time to Apogee	54.1 s
Optimum Delay	36.5 s
Velocity off Pad	38.9 m/s
Max Velocity	754 m/s
Velocity at Deployment	N/A
Landing Velocity	210 m/s

Motor	Avg Thrust	Burn Time	Max Thrust	Total Impulse	Thrust to Wt	Motor Wt	Size
5880M 1160-P	1154 N	5.1 s	1361 N	5886 Ns	12.52:1	3454 g	75/757 mm
8429M 2020-P	2021 N	4.17 s	2649 N	8429 Ns	10.71:1	4349 g	75/893 mm
Total:				14314 Ns	10.71:1	7803 g	

Drogue Only

Altitude	14468 m
Flight Time	775 s
Time to Apogee	54.1 s
Optimum Delay	36.5 s
Velocity off Pad	38.9 m/s
Max Velocity	754 m/s
Velocity at Deployment	78.2 m/s
Landing Velocity	18 m/s

Motor	Avg Thrust	Burn Time	Max Thrust	Total Impulse	Thrust to Wt	Motor Wt	Size
5880M 1160-P	1154 N	5.1 s	1361 N	5886 Ns	12.52:1	3454 g	75/757 mm
8429M 2020-P	2021 N	4.17 s	2649 N	8429 Ns	10.71:1	4349 g	75/893 mm
Total:				14314 Ns	10.71:1	7803 g	

Parts Detail

Stage: Sustainer

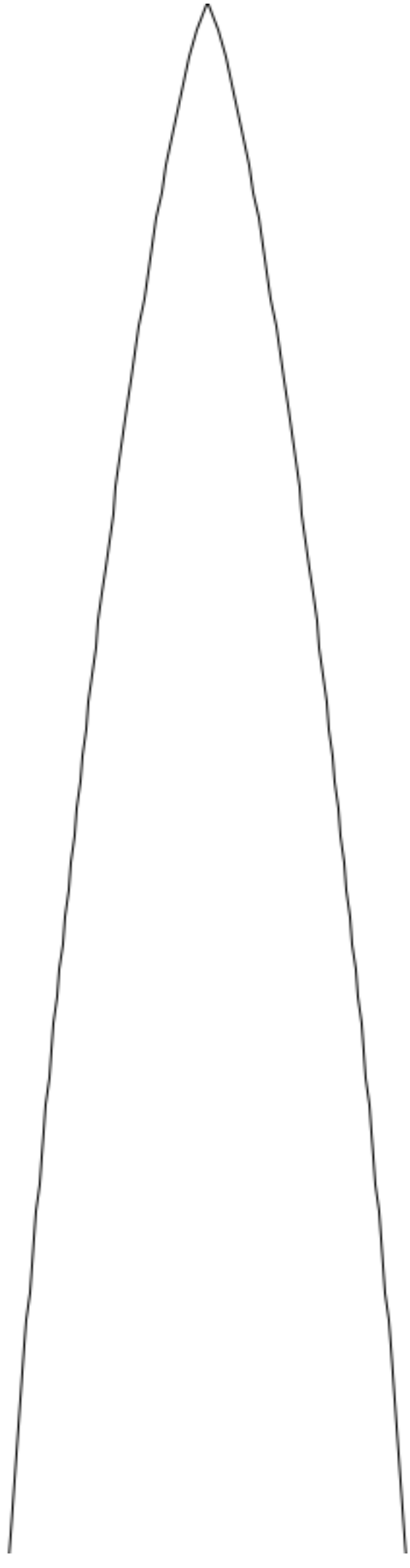
	Nose Cone	Fiberglass (1.85 g/cm³)	Haack series	Len: 44 cm	Mass: 247 g
	Nose Tip Assembly		Dia$_{out}$ 2.5 cm		Mass: 180 g
	Shock Cord	Kevlar 12-strand (10 mm, 3/8 in) (59.5 g/m)		Len: 1000 cm	Mass: 100 g
	Updated Main	Ripstop nylon (67 g/m²)	Dia$_{out}$ 170 cm	Len: 12.7 cm	Mass: 191 g
	Shroud Lines	Elastic cord (round 2 mm, 1/16 in) (1.8 g/m)	Lines: 16	Len: 255 cm	
	Shock Cord	Kevlar 12-strand (11 mm, 7/16 in) (74.4 g/m)		Len: 1000 cm	Mass: 100 g
	Swivel		Dia$_{out}$ 2.5 cm		Mass: 63.5 g
	Nose Cone Straight Length	Carbon (0.81 g/cm³)	Dia$_{in}$ 8.4 cm Dia$_{out}$ 8.8 cm	Len: 17.8 cm	Mass: 137 g
	Av Bay + Recovery	Fiberglass (1.85 g/cm³)	Dia$_{in}$ 8.4 cm Dia$_{out}$ 8.8 cm	Len: 12.4 cm	Mass: 457 g
	Tube Coupler	Fiberglass (1.85 g/cm³)	Dia$_{in}$ 7.9 cm Dia$_{out}$ 8.4 cm	Len: 38.5 cm	Mass: 0 g
	Swivel		Dia$_{out}$ 0.508 cm		Mass: 63.5 g
	Bulkhead	Fiberglass (1.85 g/cm³)	Dia$_{out}$ 8.4 cm	Len: 0.476 cm	Mass: 30 g
	Ave Bay Components + Service Module		Dia$_{out}$ 2.5 cm		Mass: 528 g
	Recovery Components		Dia$_{out}$ 2.5 cm		Mass: 220 g
	Bulkhead	Fiberglass (1.85 g/cm³)	Dia$_{out}$ 8.4 cm	Len: 0.476 cm	Mass: 30 g
	Body Tube	Carbon fiber (1.78 g/cm³)	Dia$_{in}$ 8.4 cm Dia$_{out}$ 8.8 cm	Len: 96.5 cm	Mass: 728 g
	Swivel		Dia$_{out}$ 0.508 cm		Mass: 63.5 g
	Shock Cord	Kevlar 12-strand (10 mm, 3/8 in) (59.5 g/m)		Len: 250 cm	Mass: 58 g
	Updated Drogue	Ripstop nylon (67 g/m²)	Dia$_{out}$ 81.3 cm	Len: 7.34 cm	Mass: 90 g
	Shroud Lines	Spectra #200 [Round 1.5 mm, 1/16 in] (0.656 g/m)	Lines: 8	Len: 139 cm	

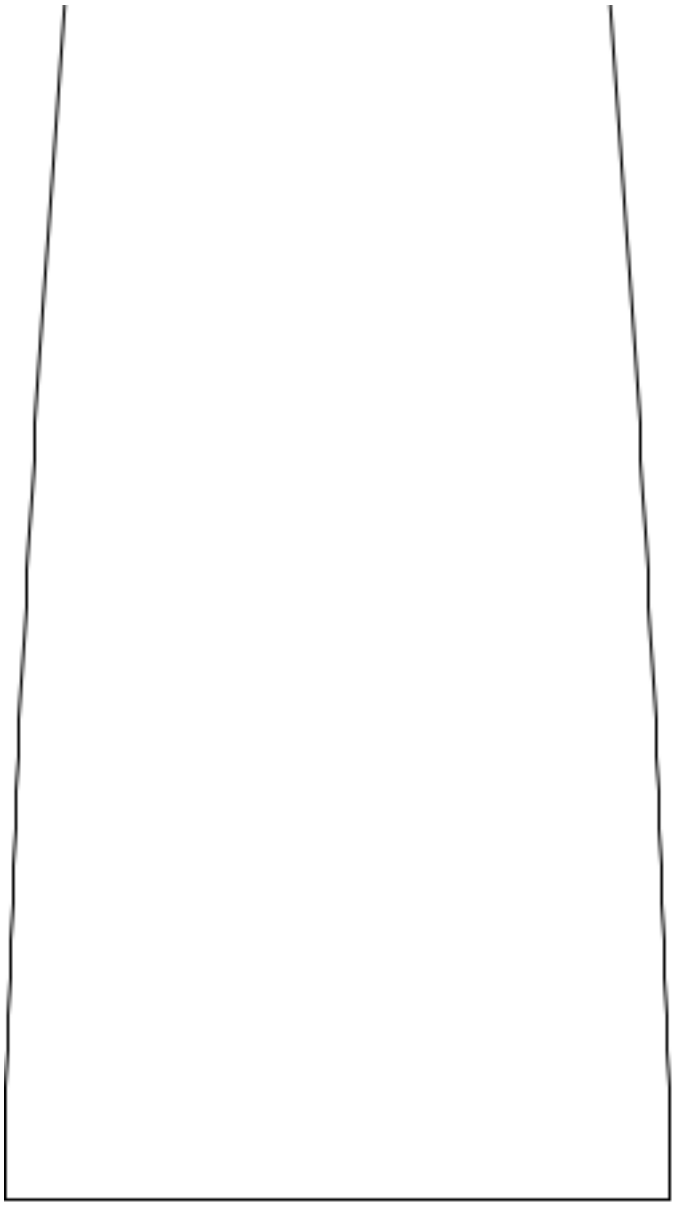
	Shock Cord	Elastic cord (round 2 mm, 1/16 in) (1.8 g/m)		Len: 250 cm	Mass: 58 g
	Swivel		Dia_{out} 0.508 cm		Mass: 63.5 g
	Motor Retention Plate + Attachment Nut	Cardboard (0.68 g/cm³)	Dia_{out} 8.4 cm	Len: 0.475 cm	Mass: 55 g
	Thrust Transfer Coupler	Carbon fiber (1.78 g/cm³)	Dia_{in} 8.4 cm Dia_{out} 8.4 cm	Len: 4.5 cm	Mass: 45 g
	Thrust Transfer Plate	Carbon fiber (1.78 g/cm³)	Dia_{in} 6.6 cm Dia_{out} 8.4 cm	Len: 0.476 cm	Mass: 9.8 g
	Centering Ring + Epoxy	Fiberglass (1.85 g/cm³)	Dia_{in} 7.65 cm Dia_{out} 8.4 cm	Len: 0.476 cm	Mass: 10 g
	Centering Ring + Epoxy	Fiberglass (1.85 g/cm³)	Dia_{in} 7.65 cm Dia_{out} 8.4 cm	Len: 0.476 cm	Mass: 10 g
	Freeform Fin Set (3)	Carbon fiber (1.78 g/cm³)	Thick: 0.44 cm		Mass: 78 g
	Centering Ring + Epoxy	Cardboard (0.68 g/cm³)	Dia_{in} 0 cm Dia_{out} 8.4 cm	Len: 0.2 cm	Mass: 10 g
	Tip-to-Tip + Fin Fillets		Dia_{out} 2.5 cm		Mass: 58 g
	Fin Leading Edge		Dia_{out} 2.5 cm		Mass: 14 g

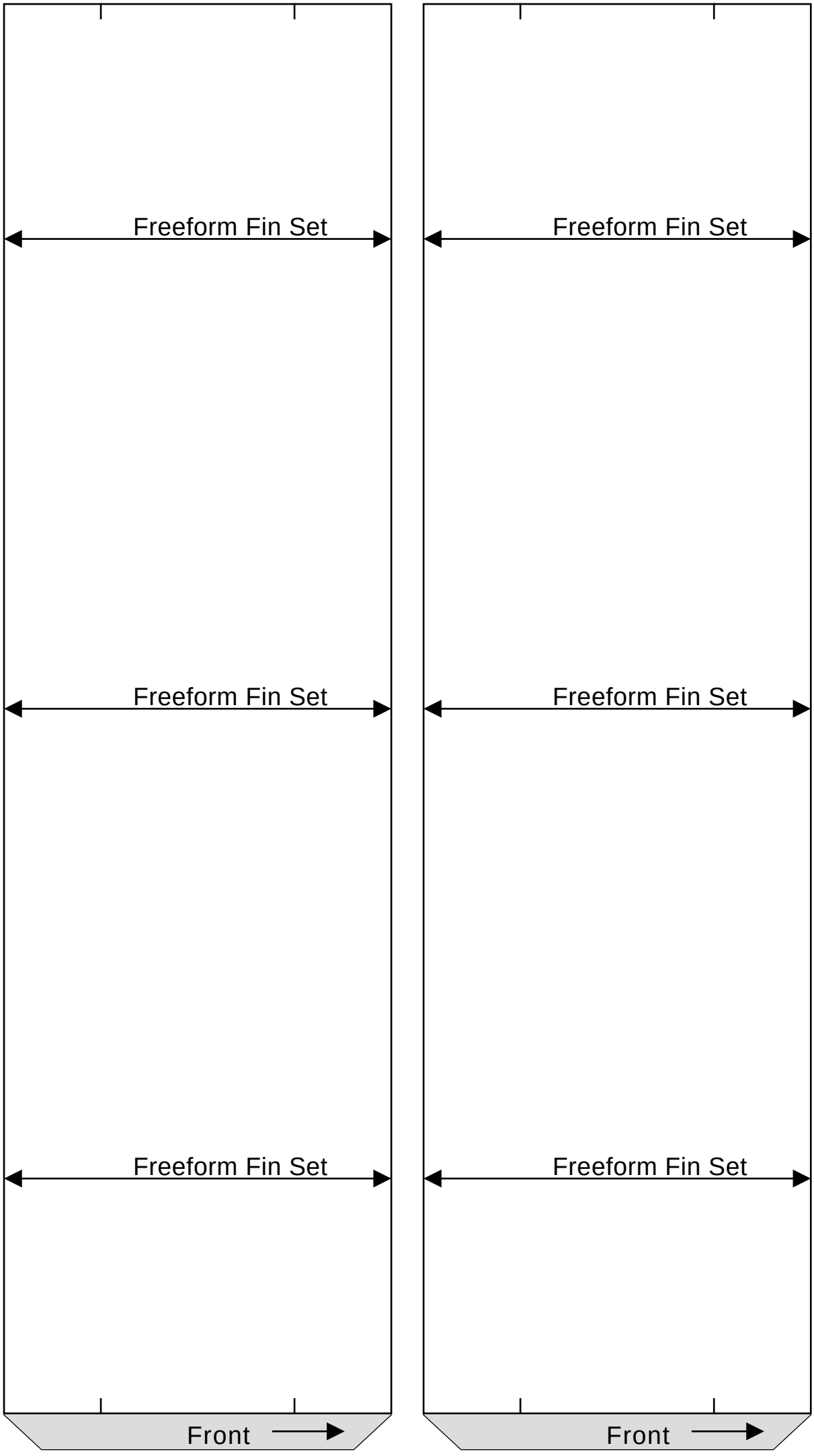
Stage: Booster

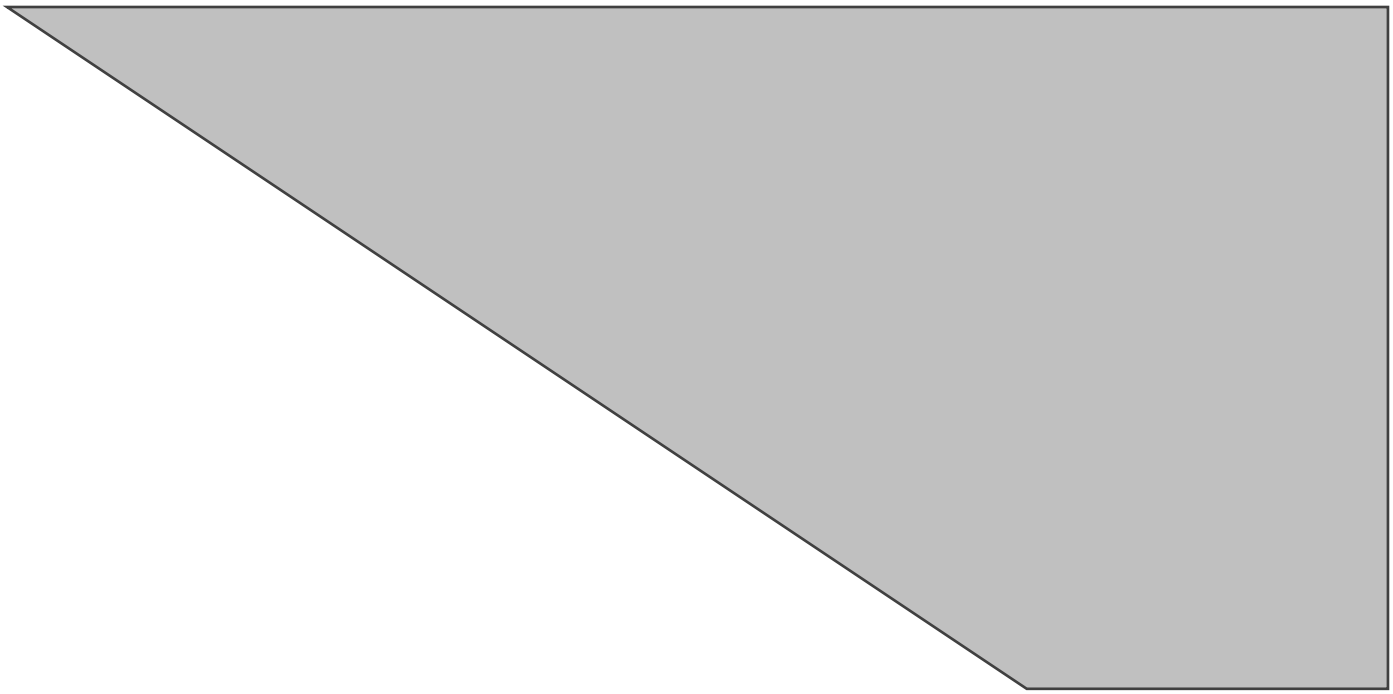
	Av. Bay	Fiberglass (1.85 g/cm³)	Dia_{in} 8.4 cm Dia_{out} 8.8 cm	Len: 10.7 cm	Mass: 459 g
	Bulkhead	Fiberglass (1.85 g/cm³)	Dia_{out} 8.4 cm	Len: 0.476 cm	Mass: 30 g
	Tube Coupler	Fiberglass (1.85 g/cm³)	Dia_{in} 7.55 cm Dia_{out} 8.4 cm	Len: 36.5 cm	Mass: 0 g
	Ave Bay Components + Service Module		Dia_{out} 2.5 cm		Mass: 301 g
	RECOVERY COMPONENTS		Dia_{out} 2.5 cm		Mass: 146 g
	Bulkhead	Fiberglass (1.85 g/cm³)	Dia_{out} 8.4 cm	Len: 0.476 cm	Mass: 30 g
	Swivel		Dia_{out} 0.635 cm		Mass: 63.5 g
	Recovery Bay + Main Tube	Carbon fiber (1.78 g/cm³)	Dia_{in} 8.4 cm Dia_{out} 8.8 cm	Len: 118 cm	Mass: 887 g
	Shock Cord	Elastic cord (round 2 mm, 1/16 in) (1.8 g/m)		Len: 1000 cm	Mass: 100 g
	Updated Main	Ripstop nylon (67 g/m²)	Dia_{out} 137 cm	Len: 5.9 cm	Mass: 110 g
	Shroud Lines	Elastic cord (round 2 mm, 1/16 in) (1.8 g/m)	Lines: 16	Len: 190 cm	

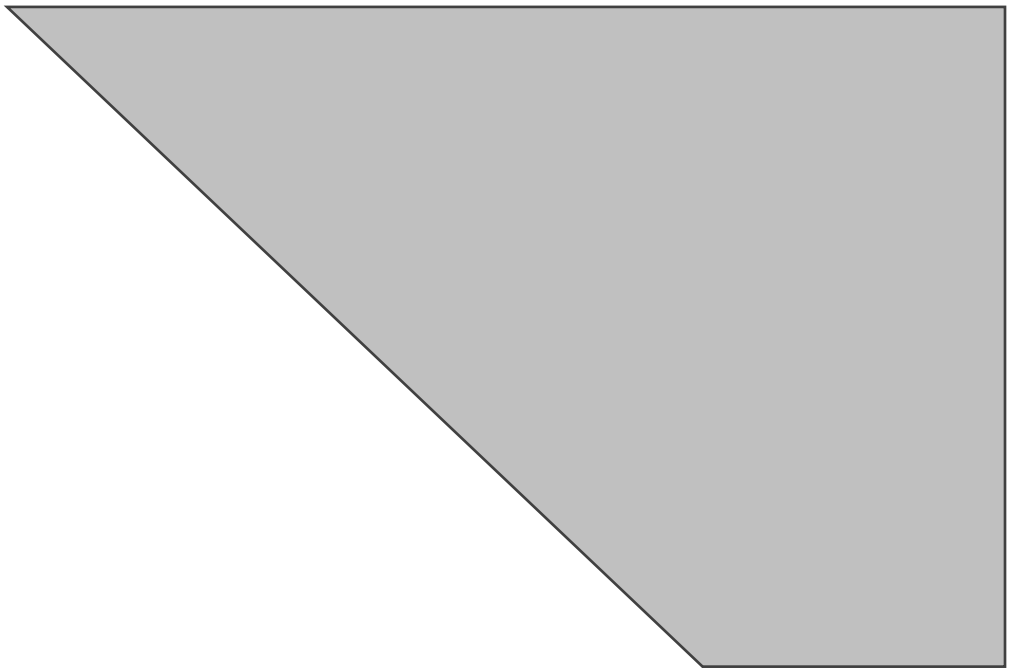
	Shock Cord	Elastic cord (round 2 mm, 1/16 in) (1.8 g/m)		Len: 1000 cm	Mass: 100 g
	Updated Drogue	Ripstop nylon, lightweight, 3 mil (35.7 g/m²)	Dia_{out} 71.1 cm	Len: 5.84 cm	Mass: 90 g
	Shroud Lines	Spectra #200 [Round 1.5 mm, 1/16 in] (0.656 g/m)	Lines: 8	Len: 122 cm	
	Shock Cord	Elastic cord (round 2 mm, 1/16 in) (1.8 g/m)		Len: 250 cm	Mass: 58 g
	Swivel		Dia_{out} 0.635 cm		Mass: 63.5 g
	Motor Retention Plate + Attachment Nut	Cardboard (0.68 g/cm³)	Dia_{out} 8.4 cm	Len: 0.475 cm	Mass: 55 g
	Thrust Transfer Coupler	Carbon fiber (1.78 g/cm³)	Dia_{in} 8.4 cm Dia_{out} 8.4 cm	Len: 4.5 cm	Mass: 45.4 g
	Thrust Transfer Plate	Carbon fiber (1.78 g/cm³)	Dia_{in} 6.6 cm Dia_{out} 8.4 cm	Len: 0.476 cm	Mass: 9.8 g
	Centering Ring + Epoxy	Fiberglass (1.85 g/cm³)	Dia_{in} 7.65 cm Dia_{out} 8.4 cm	Len: 0.476 cm	Mass: 10 g
	Centering Ring + Epoxy	Fiberglass (1.85 g/cm³)	Dia_{in} 7.65 cm Dia_{out} 8.4 cm	Len: 0.476 cm	Mass: 10 g
	Freeform Fin Set (3)	Carbon fiber (1.78 g/cm³)	Thick: 0.44 cm		Mass: 114 g
	Tip-to-Tip + Fin FIllet Mass		Dia_{out} 2.5 cm		Mass: 135 g

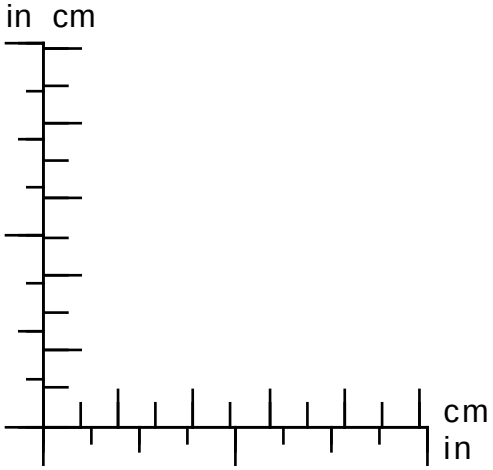





Freeform Fin Set
Freeform Fin Set
Freeform Fin Set
Freeform Fin Set
Freeform Fin Set
Freeform Fin Set
Front
Front






in cm
cm
in

